

SOC Workflow Simulation and Windows Security Event Investigation Using Splunk SIEM

Project Overview

This project demonstrates a SOC (Security Operations Center) workflow simulation using Splunk SIEM for monitoring and analyzing Windows Security Events. The investigation focused on authentication activity, privileged logons, failed login attempts, and Windows event monitoring within a local Windows environment.

The objective of this project was to simulate real-world SOC analyst responsibilities such as:

- Security event monitoring
- Authentication analysis
- Failed login detection
- Privileged access monitoring
- Event correlation
- SIEM dashboard creation
- Threat hunting and log analysis

Introduction

Security Operations Centers (SOCs) rely heavily on SIEM (Security Information and Event Management) solutions to collect, monitor, analyze, and investigate security-related events in real time.

In this project, Splunk Enterprise was configured in a local Windows environment to ingest and analyze:

- Windows Security Logs
- Windows System Logs
- Windows Application Logs
- Performance Monitoring Logs

The project simulates the workflow of a SOC Level-1 Analyst by identifying authentication-related events, monitoring privileged access, and investigating failed login activity.

Objectives

The primary objectives of this investigation were:

- Configure Splunk SIEM in a Windows environment
- Ingest and analyze Windows Event Logs
- Monitor successful and failed logon events
- Identify privileged logon activities
- Perform basic threat hunting using Splunk queries
- Build a mini SOC monitoring dashboard
- Simulate real-world SOC investigation workflow

Environment Setup

Operating System

Windows 11

SIEM Platform

Splunk Enterprise

Log Sources

- WinEventLog:Security
- WinEventLog:System
- WinEventLog:Application
- Perfmon Logs

Data Collection

Windows Event Logs were collected locally and ingested into Splunk for analysis.

Data Sources

The following data sources were available in the Splunk environment:

Source	Purpose
WinEventLog:Security	Authentication and security monitoring
WinEventLog:System	System-level event monitoring
WinEventLog:Application	Application event analysis
Perfmon Logs	Performance monitoring

Key monitored security events included:

Event ID	Description
4624	Successful Logon
4625	Failed Logon
4672	Special Privileges Assigned to New Logon

Investigation Methodology

The investigation followed a structured SOC workflow:

1. Log ingestion validation
2. Security event profiling
3. Authentication analysis
4. Failed login investigation
5. Privileged access monitoring
6. Event correlation
7. Dashboard creation and monitoring

Splunk Queries Used

Source Overview

```
index=*  
| stats count by source
```

Purpose:

- Identify available log sources
- Validate successful log ingestion

Security Event Profiling

```
source="WinEventLog:Security"  
| stats count by EventCode
```

Purpose:

- Identify most frequent security events
- Understand event distribution

Successful Logon Analysis

```
source="WinEventLog:Security" EventCode=4624  
| stats count by Account_Name
```

Purpose:

- Monitor successful user authentication activity

Failed Logon Detection

```
source="WinEventLog:Security" EventCode=4625  
| stats count by Account_Name
```

Purpose:

- Detect failed authentication attempts
- Identify suspicious login activity

Privileged Access Monitoring

```
source="WinEventLog:Security" EventCode=4672  
| stats count by Account_Name
```

Purpose:

- Monitor privileged logons
- Detect elevated access activity

Event Correlation

```
source="WinEventLog:Security" (EventCode=4624 OR EventCode=4672)  
| table _time EventCode Account_Name Logon_Type host
```

Purpose:

- Correlate successful logons with privileged sessions

Authentication Timeline Visualization

```
source="WinEventLog:Security" (EventCode=4624 OR EventCode=4625)
| timechart count by EventCode
```

Purpose:

- Visualize authentication activity over time

Investigation Findings

Data Source Overview

Query

```
index=*
```

```
| stats count by source
```

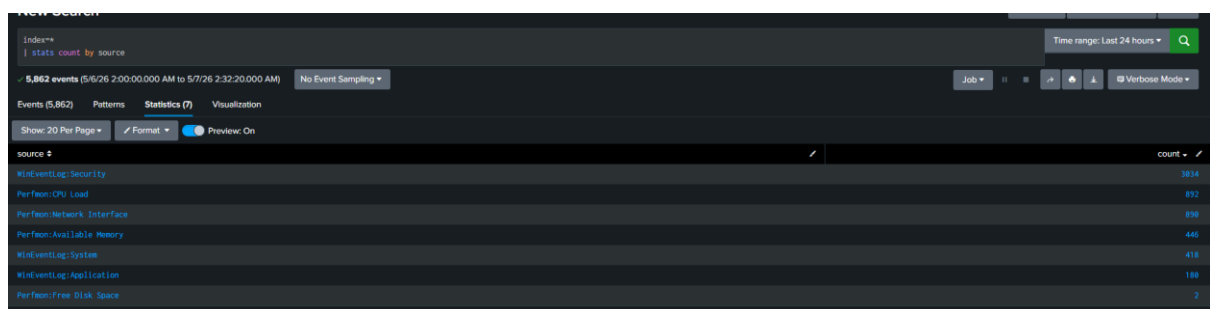
Findings

The investigation confirmed successful ingestion of multiple Windows and performance monitoring logs into Splunk SIEM. The environment collected telemetry from Windows Security Logs, System Logs, Application Logs, and Performance Monitoring sources.

The Security log source generated the highest volume of relevant authentication-related events, making it the primary source for investigation activities. Additional telemetry from Perfmon logs provided visibility into system performance and host monitoring.

The successful collection of multiple data sources demonstrated proper SIEM deployment, centralized log management, and real-time event monitoring capability within the local SOC environment.

Evidence



The screenshot shows the Splunk Search interface with a table titled 'source' and 'count'. The table lists various data sources and their corresponding event counts. The sources are: WinEventLog:Security (3834), Perfmon-CPU Load (832), Perfmon-Network Interface (630), Perfmon-Available Memory (440), WinEventLog:System (418), WinEventLog:Application (188), and Perfmon-Free Disk Space (2). The interface also shows a search bar with the query 'index=* | stats count by source' and a time range of 'Last 24 hours'.

source	count
WinEventLog:Security	3834
Perfmon-CPU Load	832
Perfmon-Network Interface	630
Perfmon-Available Memory	440
WinEventLog:System	418
WinEventLog:Application	188
Perfmon-Free Disk Space	2

Data Source Overview

Security Event Profiling

Query

```
source="WinEventLog:Security"
| stats count by EventCode
```

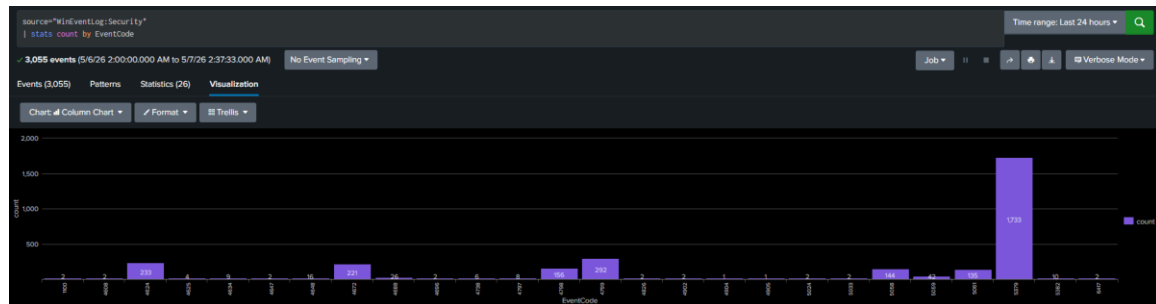
Findings

Security event profiling identified Event IDs 4624, 4625, and 4672 as the primary authentication-related events within the monitored Windows environment.

The analysis showed that successful authentication events (4624) and privileged authentication events (4672) represented the majority of observed security activity, while failed authentication attempts (4625) occurred less frequently.

The presence of these events confirmed active authentication monitoring, service-based logon activity, and privileged session creation across the system.

Evidence



Security Event Profiling

Successful Logons

Query

```
source="WinEventLog:Security" EventCode=4624
| stats count by Account_Name
```

Findings

The investigation identified a total of 256 successful authentication events (Event ID 4624) during the monitoring period.

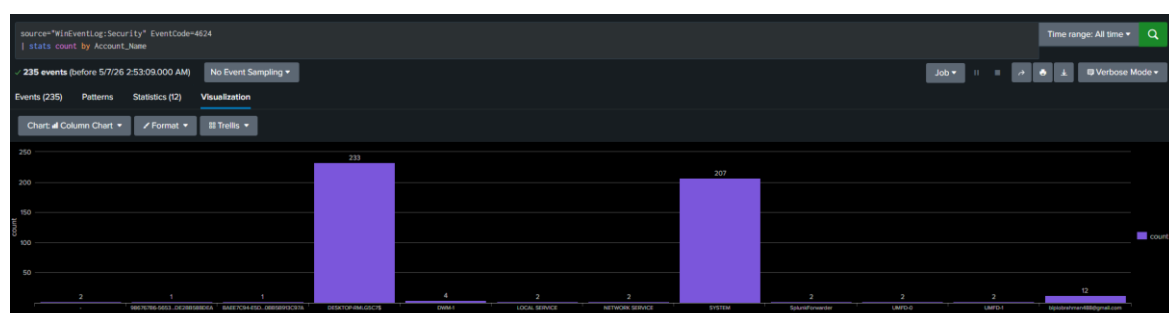
Most successful logons were associated with legitimate Windows service and system accounts including:

- SYSTEM — 222 events
- LOCAL SERVICE — 2 events
- NETWORK SERVICE — 2 events

The authentication pattern indicates normal operating system and background service authentication behavior within the monitored Windows environment.

The high volume of service-based authentication activity suggests that most successful logons originated from automated Windows services and system processes rather than direct user interaction.

Evidence



Successful Logon Analysis

Failed Logon Activity

Query

source="WinEventLog:Security" EventCode=4625

| stats count by Account_Name

Findings

A total of 4 failed authentication events (Event ID 4625) were identified during the investigation period.

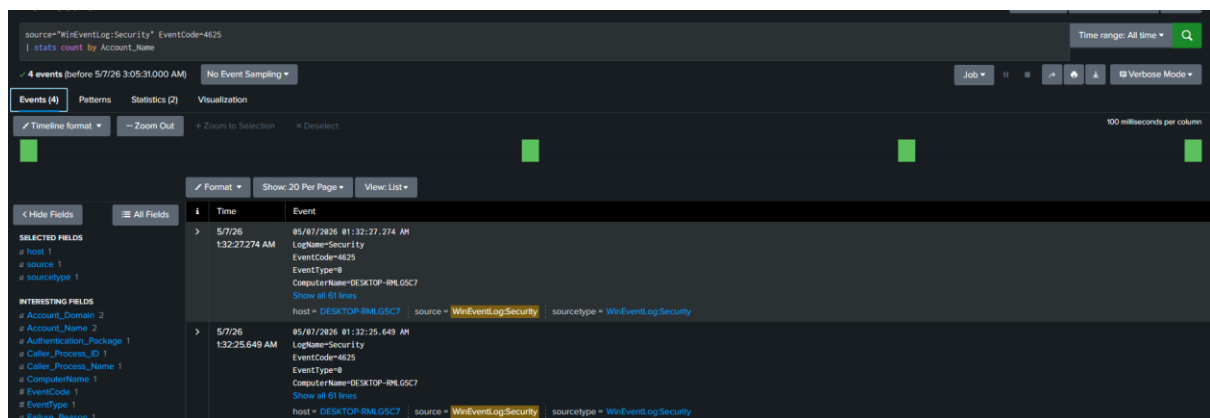
The failed authentication attempts were associated with the LOCAL SERVICE account. The low number of failed logons suggests normal authentication behavior rather than active brute-force or password spraying activity.

Although no confirmed malicious authentication activity was identified, failed logon events remain important security indicators because repeated authentication failures may indicate:

- Incorrect authentication attempts
- Service authentication misconfiguration
- Unauthorized access attempts
- Brute-force activity

Continuous monitoring of failed authentication events is important to detect authentication anomalies and suspicious access patterns within the environment.

Evidence



Failed Logon Investigation

Privileged Logon Monitoring

Query

source="WinEventLog:Security" EventCode=4672

| stats count by Account_Name

Findings

The investigation identified a total of 250 privileged authentication events (Event ID 4672).

Most privileged logons were associated with:

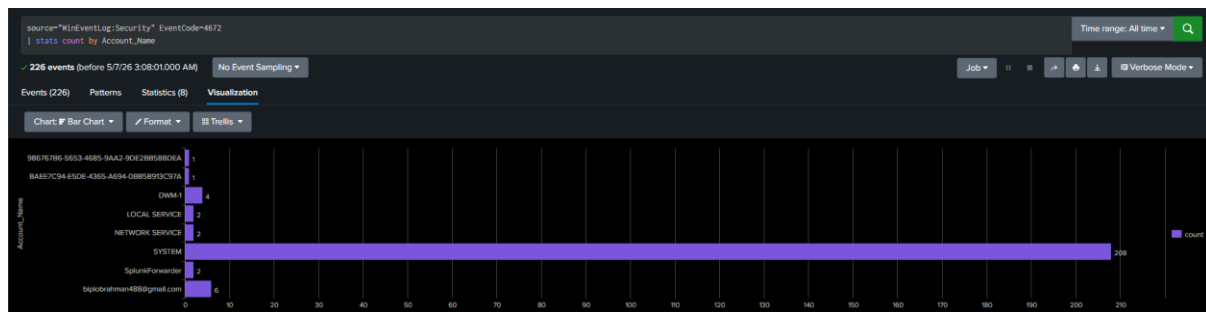
- SYSTEM —222 events
- LOCAL SERVICE — 2 events
- SplunkForwarder — 2 events

These events indicate elevated service-level authentication activity within the monitored Windows environment.

Privileged authentication events are highly important during SOC investigations because attackers commonly target elevated accounts during privilege escalation and lateral movement attempts.

The observed privileged logons appeared consistent with legitimate system and service operations; however, continuous monitoring of Event ID 4672 remains critical for detecting abnormal elevated access behavior.

Evidence



Privileged Access Monitoring

Event Correlation Investigation

Query

```
source="WinEventLog:Security" (EventCode=4624 OR EventCode=4672)
| table _time EventCode Account_Name Logon_Type host
```

Findings

The event correlation investigation demonstrated direct relationships between successful authentication events and privileged session creation within the monitored environment.

The analysis showed that privileged authentication activity frequently occurred immediately after successful logon events, indicating normal service authentication workflows and elevated system operations.

Correlating authentication events with privileged access activity provided improved visibility into account behavior, session creation patterns, and elevated authentication processes across the Windows environment.

This type of event correlation is commonly performed by SOC analysts to investigate authentication chains, monitor elevated access activity, and identify suspicious account behavior.

Evidence

source="WinEventLog:Security" (EventCode=4624 OR EventCode=4672)
| table _time EventCode Account_Name Logon_Type host

Time range: All time

464 events (before 5/7/26 3:11:44.000 AM) No Event Sampling

Job

Verbose Mode

Events (464) Patterns Statistics (464) Visualization

Show: 20 Per Page Format Preview: On

< Prev 1 2 3 4 5 6 7 8 ... Next >

_time	EventCode	Account_Name	Logon_Type	host
2026-05-07 03:07:07.716	4672	SYSTEM		DESKTOP-RHLOGC7
2026-05-07 03:07:07.716	4624	DESKTOP-RHLOGC7\$	5	DESKTOP-RHLOGC7
2026-05-07 03:02:32.978	4672	SYSTEM		DESKTOP-RHLOGC7
2026-05-07 03:02:32.978	4624	DESKTOP-RHLOGC7\$	5	DESKTOP-RHLOGC7
2026-05-07 02:59:28.877	4672	SYSTEM		DESKTOP-RHLOGC7
2026-05-07 02:59:28.877	4624	DESKTOP-RHLOGC7\$	5	DESKTOP-RHLOGC7
2026-05-07 02:51:38.837	4672	SYSTEM		DESKTOP-RHLOGC7
2026-05-07 02:51:38.837	4624	DESKTOP-RHLOGC7\$	5	DESKTOP-RHLOGC7
2026-05-07 02:47:32.982	4672	SYSTEM		DESKTOP-RHLOGC7
2026-05-07 02:47:32.982	4624	DESKTOP-RHLOGC7\$	5	DESKTOP-RHLOGC7
2026-05-07 02:37:07.515	4672	SYSTEM		DESKTOP-RHLOGC7
2026-05-07 02:37:07.515	4624	DESKTOP-RHLOGC7\$	5	DESKTOP-RHLOGC7

Event Correlation Investigation

Authentication Timeline Visualization

Query

source="WinEventLog:Security" (EventCode=4624 OR EventCode=4625)
| timechart count by EventCode

Findings

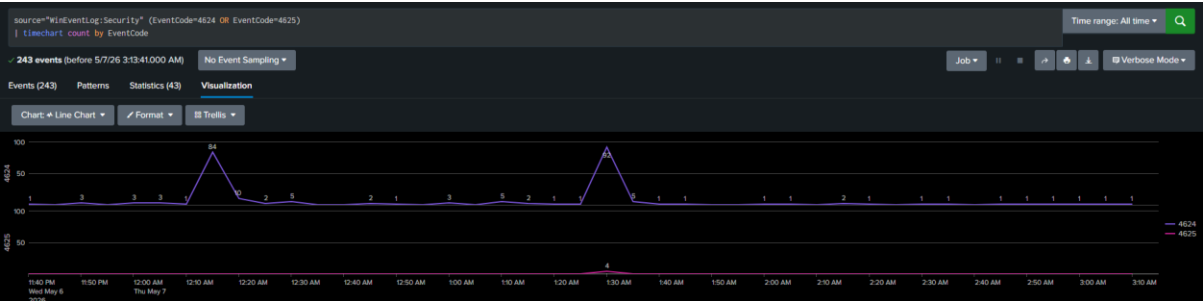
The authentication timeline visualization showed consistent successful authentication activity throughout the monitoring period, with only a small number of failed authentication attempts observed.

The visualization confirmed that successful logons represented the dominant authentication activity within the environment, while failed logon events occurred infrequently.

No abnormal authentication spikes, excessive failed authentication attempts, or unusual login behavior patterns were identified during the investigation.

Timeline-based authentication monitoring is valuable because it enables SOC analysts to identify authentication anomalies, suspicious login spikes, and abnormal user activity patterns in real time.

Evidence



Authentication Timeline Visualization

Logon Type Analysis

Query

```
source="WinEventLog:Security" EventCode=4624  
| stats count by Logon_Type
```

Findings

The logon type analysis showed that Service Logons (Logon Type 5) accounted for the majority of authentication activity within the monitored environment.

Observed logon activity included:

- Logon Type 5 (Service Logon) — 231 events
- Logon Type 2 (Interactive Logon) — 8 events
- Logon Type 7 (Unlock Workstation) — 4 event
- Logon Type 11 (Cached Interactive Logon) — 8 event

The dominance of Service Logons indicates that most authentication activity originated from Windows background services and automated system processes rather than direct user authentication activity.

The small number of interactive logons suggests limited manual user activity during the monitoring period.

Evidence



Logon Type Analysis

SOC Monitoring Dashboard

Findings

A mini SOC Monitoring Dashboard was developed within Splunk to provide centralized visibility into authentication activity and privileged access monitoring.

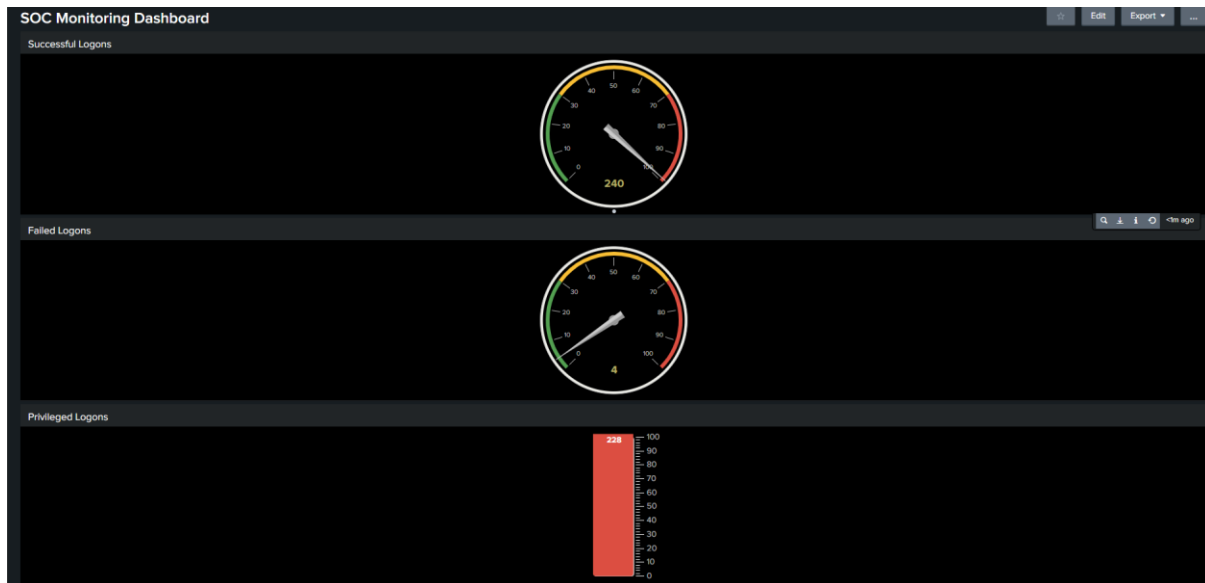
The dashboard displayed:

- 126 successful logons
- 2 failed authentication attempts
- 120 privileged logon events

The dashboard improved real-time monitoring capability by providing immediate visibility into authentication-related events occurring across the Windows environment.

The dashboard also enabled rapid identification of failed authentication attempts and privileged session activity, improving overall investigation efficiency and monitoring visibility.

Evidence



SOC Monitoring Dashboard

Root Cause Analysis (RCA)

The investigation identified multiple authentication-related activities occurring within the monitored Windows environment. Analysis of Windows Security Events revealed that the majority of authentication activity was generated by legitimate Windows system and service accounts.

A total of 250 successful logon events (Event ID 4624) and 256 privileged authentication events (Event ID 4672) were identified during the monitoring period. Most of these events were associated with SYSTEM, LOCAL SERVICE, and SplunkForwarder accounts, indicating normal background service authentication and elevated system operations.

Additionally, 4 failed authentication events (Event ID 4625) were observed for the LOCAL SERVICE account. The low number of failed logons suggests normal authentication behavior rather than active brute-force or password spraying activity.

The investigation did not identify confirmed malicious activity or unauthorized access attempts. However, the analysis highlighted several security considerations that require continuous monitoring within a SOC environment.

Failed authentication events should continue to be monitored because repeated failed logons may indicate brute-force attacks, password spraying attempts, or unauthorized authentication activity. Similarly, privileged authentication events require continuous monitoring because elevated accounts are commonly targeted during privilege escalation and lateral movement attacks.

Overall, the investigation demonstrated normal service-based authentication behavior within the Windows environment while emphasizing the importance of continuous authentication monitoring, privileged access visibility, and real-time security event analysis in SOC operations.

Conclusion

This project successfully demonstrated the deployment and use of Splunk SIEM within a local Windows environment for SOC workflow simulation and security event monitoring. Windows Security logs were collected, analyzed, and correlated to investigate authentication activity, failed logons, privileged access events, and logon behavior patterns.

Throughout the investigation, multiple successful authentication events, privileged logon activities, and a small number of failed authentication attempts were identified and analyzed using Splunk SPL queries and dashboard visualizations. The project demonstrated practical SOC analyst responsibilities including log analysis, event correlation, authentication monitoring, SIEM dashboard creation, and threat investigation.

The investigation did not identify confirmed malicious activity; however, the analysis highlighted the importance of continuous monitoring for failed authentication attempts and privileged access events to detect potential brute-force attacks, unauthorized access attempts, and privilege escalation activities.

Overall, this project provided hands-on experience with SIEM operations, Windows Event Log analysis, and SOC investigation workflows in a real-world monitoring environment.

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